

ORIGINAL RESEARCH PAPER

Single sample per person face recognition algorithm based on the robust prototype dictionary and robust variation dictionary construction

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Abstract

Single sample per person (SSPP) face recognition uses only a single face image of each subject in the gallery set to recognize the probe sample. Since single sample cannot provide intra-class variation information, the matching accuracy of the gallery faces with the faces captured in unconstrained video is usually low. Recently, in order to improve the accuracy, the sparse representation-based classification (SRC) technology has been extended to generic learning method, which uses prototype and variation dictionary (P+V) model for face recognition. Because the inter-class scatter between atoms in prototype dictionary is not big enough, and the intra-class scatter in constructed variation dictionary (such as posture and expression) is not rich enough, the robustness of P+V model for SSPP face recognition is poor. To solve this problem, a robust prototype dictionary and robust variation dictionary construction (RPRV) method is proposed. First, a set of atoms is obtained by dictionary learning method using gallery images and generic images. Second, some effective atoms are selected by the proposed function index method. Finally, the robust prototype dictionary (RP) and the robust variation dictionary (RV) are represented linearly using these effective atoms, respectively. The face recognition is performed according to the proposed RP+RV model. Experiment results using public datasets show that the proposed RPRV has strong robustness for the face captured under the unconstrained environment. Comparison results show that the proposed RPRV method outperform state-of-the-art SRC-based methods for SSPP face recognition.

1 | INTRODUCTION

Face recognition is an important research field in computer science and many application scenarios over the past decades [1–6]. In many practical face recognition systems, for example, law enforcement, entrance monitoring, intelligent video surveillance, criminal identification, the gallery set is typically designed using only one single sample per person (SSPP) [7] because of the limited storage and privacy policy. What makes the situation even worse is that the probe samples are usually captured with video surveillance cameras under unconstrained conditions [8, 9]. Therefore, SSPP face recognition in this unconstrained environment is very challenging task due to, first, the strong intra-class variations such as illumination [10–12], pose [12, 13], facial

expression [14–16] and occlusion [17, 18], and second, the lack of gallery samples.

Up to date, four main kinds of methods for SSPP face recognition are reported, including virtual sample generation [19–21], classifier [22–24], image partitioning [25–29] and generic learning [30–40]. The virtual sample generation method uses the singular value decomposition (SVD) [19], pose and illumination variability [20], the lower-upper decomposition [21] etc. to generate multiple virtual samples from a single gallery image of the subject, so that the conventional techniques using multiple samples per subject can be applied for SSPP face recognition. Although the recognition results of these algorithms can be improved to a certain extent compared with the original single sample, the extra information generated through virtual

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samples is quite limited. Therefore, the performance improvement is quite limited.

The classifier method uses training, determination etc. to make the distance between the same class of frontal gallery sample and the probe sample close to the distance between the different class. Plichoskia et al. [22] proposed an SSPP method based on differential evolution algorithm, applied the differential evolution method to the face recognition framework for the first time, which has good performance under constrained environment. Pang et al. [23] proposed a iterative dynamic generic learning (IDGL) method to handle SSPP problem with a contaminated biometric enrolment database, which has good performance under unconstrained environment. Zhang et al. [24] proposed an SSPP method based on dissimilarity-based nearest neighbor classifier, which is effective on the AR dataset. However, the classifier training process of gallery class is time-consuming. In order to recognize in real time, this method is usually run offline after training. These recognition system cannot change the gallery class immediately under given circumstances.

The image partitioning method divides the single gallery image into several overlapping/nonoverlapping image patches, and each patch is classified independently. Then, the classification results of the probe image patches are obtained, and the final classification result of the probe image is obtained by voting or other methods. Zhu et al. [25] extended the collaborative representation based classification (CRC) [26] to image partition based method, that is, PCRC, by integrating the CRC outputs of all partitioned patches. First, the recognition results of image patches are obtained by CRC method, then the recognition result of one scale is obtained by voting method, finally, under the multi-scale, the recognition result is obtained by the weighted combination. Furthermore, Yan et al. [27] and Zhang et al. [28] proposed the manifold learning methods, first, the feature is extracted for each image patch, then the graph structure of each image is obtained using its patch features, and finally the recognition result is obtained by matching the graph structure. Adjabi et al. [29] proposed an original method based on multi-block color-binarized statistical image features to handle SSPP problem, which method has faster running speed. These methods use the similarity of patches in the same position to identify a probe patch. However, when there is pose variations or misalignments in the probe sample, the patches of the same position of the probe and gallery sample may not match well.

The generic learning method assumes that similar intra-class variations could be shared by different individuals. First, the intra-class variation information is obtained by an auxiliary generic set; this variation information is represented as the difference of gallery samples and probe samples [30, 31]. Second, the sparse coefficients between gallery samples and probe sample are obtained. Finally, the identity of the probe sample is obtained by the sparse coefficients. The generic learning method generally outperforms the other two kinds of methods. Sparse representation classification (SRC) [32] has better face recognition performance when each person has more than one sample in the gallery set. However, it is not suitable for SSPP

because of the single gallery sample. Recently, SRC has been extended to generic learning method by using prototype and variation dictionary (P+V) model for SSPP. Deng et al. [33] proposed an extended sparse representation classification (ESRC) method, by using an auxiliary intra-class variation dictionary to represent the difference between the gallery and probe images. The atoms in variation dictionary typically represent intra-class difference of samples computed from the generic set. Motivated by ESRC, Yang et al. [34] proposed the sparse variation dictionary learning (SVDL) model, a projection from the gallery set to the generic set is learned, and the sparse variation dictionary is obtained by the projection using the generic set. By further extending SVDL, Ji et al. [35] proposed a collaborative probabilistic labels (CPL) method. First, the probabilistic labels are constructed by label propagation using the gallery samples and the generic samples. Then, the adaptive variation dictionary is obtained using the group sparse coding method. Finally, the probe sample is recognized using its corresponding adaptive variation dictionary and probabilistic labels. Moreover, Deng et al. [36] presented a superposed SRC based P+V model, namely, a probe sample is considered as the sum of the prototype and the intra-class variation. In their P+V model, the prototype is constituted by the sample in the gallery set since the gallery sample is assumed to be standard and variation-free. Yang et al. [37] proposed joint and collaborative representation with local adaptive convolution feature (JCR-ACF) based on the P+V model. The local region is obtained by densely sampling and sparsely detecting facial points method, and deep features are learned by convolutional neural networks (CNN) using the images of the local region. Then probe image of local region is recognized using the deep features and a generic set. Zhu et al. [38] proposed a local generic representation (LGR) based framework for SSPP face recognition. A local prototype dictionary is built by the neighboring patches from the gallery set, while an intra-class variation dictionary is built by using a generic set to predict the facial variation. Gao et al. [39] proposed a semi-supervised sparse representation based classification (S^3 RC), the identity of the gallery sample of the minimum deviation with probe sample is the recognition result. Gao et al. [40] proposed a synergistic generic learning (SGL) method under the P+V model framework. Although SGL achieves a promising performance for SSPP face recognition, there still exist two problems. First, the neutral samples in prototype dictionary are difficult to identify, since the training dataset in unconstrained environment does not contain neutral samples of each subject. Therefore, SGL will not be well fit for this unconstrained case. Second, the variation information provided by the constructed variation dictionary is limited, resulting in the limited performance improvement.

SSPP face recognition usually contains only a single positive neutral face image in the gallery set, and the probe samples in the unconstrained environment are affected by illumination changes, expression changes, occlusion and other factors. For the SSPP, the probe sample is difficult to match with the gallery samples because the diversity of samples in the gallery set does not exist. The prototype dictionary plus variation dictionary (P+V) model [36] can provide a solution for this

problem. In [36], each generic sample or the difference between each generic sample and the mean of all the generic samples of this class is regarded as an atom, all atoms are constructed as the variation dictionary, this dictionary represents variation information of the probe sample. At the same time, the single sample in the gallery set is regarded as an atom in the prototype dictionary, and this prototype dictionary represents the class information. The probe sample is reconstructed using the prototype dictionary and the variation dictionary, and the sparse coefficients of the atoms are obtained. Finally, the coefficients of the atoms in the variation dictionary represents the variation degree of the probe sample, the predicted class of the probe sample is obtained by using the coefficients of the atoms in the prototype dictionary. In P+V model, the single gallery sample shares the variation information in the auxiliary generic set when the probe sample is reconstructed, although it solves the SSPP problem of not enough diversity of samples in the gallery set to some extent, there are still two problems for the construction of variation dictionary in P+V model: (1) The constructed variation dictionary contains some inter-class information, which makes the recognition degrade; (2) The prototype dictionary constructed using only the gallery samples cannot improve the inter-class discriminative ability, which limits the recognition accuracy improvement.

To deal with the above two problems, under the framework of the P+V model [36], we propose a robust prototype dictionary and robust variation dictionary construction (RPRV) method to handle the SSPP face recognition problem in unconstrained environment. First, aiming at the problem that atoms in the variation dictionary have some inter-class scatter information, we propose a robust variation dictionary construction method to reduce the inter-class scatter. A ratio of inter-class scatter to intra-class scatter is defined first, and the atoms with the smaller ratio are selected to construct robust variation dictionary, which reduces the inter-class scatter in the variation dictionary. Second, aiming at the problem that the discriminative ability of atoms in prototype dictionary is insufficient, we propose a robust prototype dictionary construction method. A function of inter-class scatter is defined, then, the atoms with bigger function value are selected to construct prototype dictionary, which improves the inter-class discrimination ability. Using RPRV, we derive a robust method for SSPP to improve the recognition accuracy.

The major contributions of the proposed method are listed below:

- 1) A ratio index is proposed for the variation dictionary atoms selection, which helps to reduce the inter-class discriminative ability in the variation dictionary;
- 2) A function index is defined for the prototype dictionary atoms selection, which improves the inter-class discriminative ability in the prototype dictionary;
- 3) The proposed variation dictionary and prototype dictionary construction method is more robust in P+V SSPP recognition framework so as to achieve better recognition accuracy as compared to existing methods.

The rest of the paper is organized as follows: The proposed algorithm for SSPP face recognition is given in Section 2. The experimental results on the constrained datasets and unconstrained datasets, and the comparison with existing methods are given in Section 3. The conclusions are given in Section 4.

2 | THE PROPOSED METHOD

The sketch map of the RPRV construction method is shown in Figure 1. gallery images and generic images are combined for dictionary learning in order to construct the robust prototype dictionary (RP) and robust variation dictionary (RV). The robust prototype dictionary and robust variation dictionary are used to linearly represent a reconstructed sample with the same label as the prototype used. The label with the smallest residual error between the reconstructed sample and the probe sample is the recognition result. In this method, the enhanced robust prototype dictionary and robust variation dictionary help to improve the SSPP face recognition performance.

For the convenience of description, let $\mathbf{y} \in \mathcal{R}^d$ be a probe sample, where d denotes the dimension of a sample. Let $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_m] \in \mathcal{R}^{d \times m}$ be the gallery data matrix, where m denotes the number of the gallery subjects, each subject has only one sample. The auxiliary generic data matrix is defined as $\mathbf{A} = [\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_C] \in \mathcal{R}^{d \times n}$, where C denotes the number of the generic subjects, n denotes the number of the generic samples, and $\mathbf{A}_i \in \mathcal{R}^{d \times n_i}$ denotes that subject i contains n_i samples, $\sum_{i=1}^C n_i = n$, $\mathbf{A}_i \in \Omega_i$, Ω_i denotes class i in the data matrix.

The variation matrix is constructed by the generic samples and their class centroids as follows:

$$\hat{\mathbf{V}} = [\mathbf{A}_1 - \mathbf{c}_1 \mathbf{e}_1^T, \mathbf{A}_2 - \mathbf{c}_2 \mathbf{e}_2^T, \dots, \mathbf{A}_C - \mathbf{c}_C \mathbf{e}_C^T], \quad (1)$$

where $\mathbf{e}_i = [1, \dots, 1]^T \in \mathcal{R}^{n_i \times 1}$, $\mathbf{c}_i = \mathbf{A}_i \mathbf{e}_i / n_i$ denotes the geometric centroid of class Ω_i . Let $\mathbf{V}_i = \mathbf{A}_i - \mathbf{c}_i \mathbf{e}_i^T$ be the variation matrix of class Ω_i .

2.1 | Dictionary learning

Gallery data matrix $\mathbf{G} \in \mathcal{R}^{d \times m}$ and variation matrix $\hat{\mathbf{V}} \in \mathcal{R}^{d \times n}$ are merged into the matrix $\mathbf{X} \in \mathcal{R}^{d \times (m+n)}$ using the following Equation (2)

$$\mathbf{X} = [\mathbf{G}, \hat{\mathbf{V}}]. \quad (2)$$

The dictionary $\mathbf{D}$ is learned from the matrix $\mathbf{X}$

$$\mathbf{X} = \mathbf{D}\mathbf{W} + \mathbf{E}, \quad (3)$$

where $\mathbf{D} \in \mathcal{R}^{d \times k}$ denotes variation dictionary, k is the number of atoms in the dictionary ($k \leq (m+n) \ll d$), $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{(m+n)}] \in \mathcal{R}^{k \times (m+n)}$ is the sparse coefficient matrix, and $\mathbf{E} \in \mathcal{R}^{d \times (m+n)}$ denotes the residual matrix.

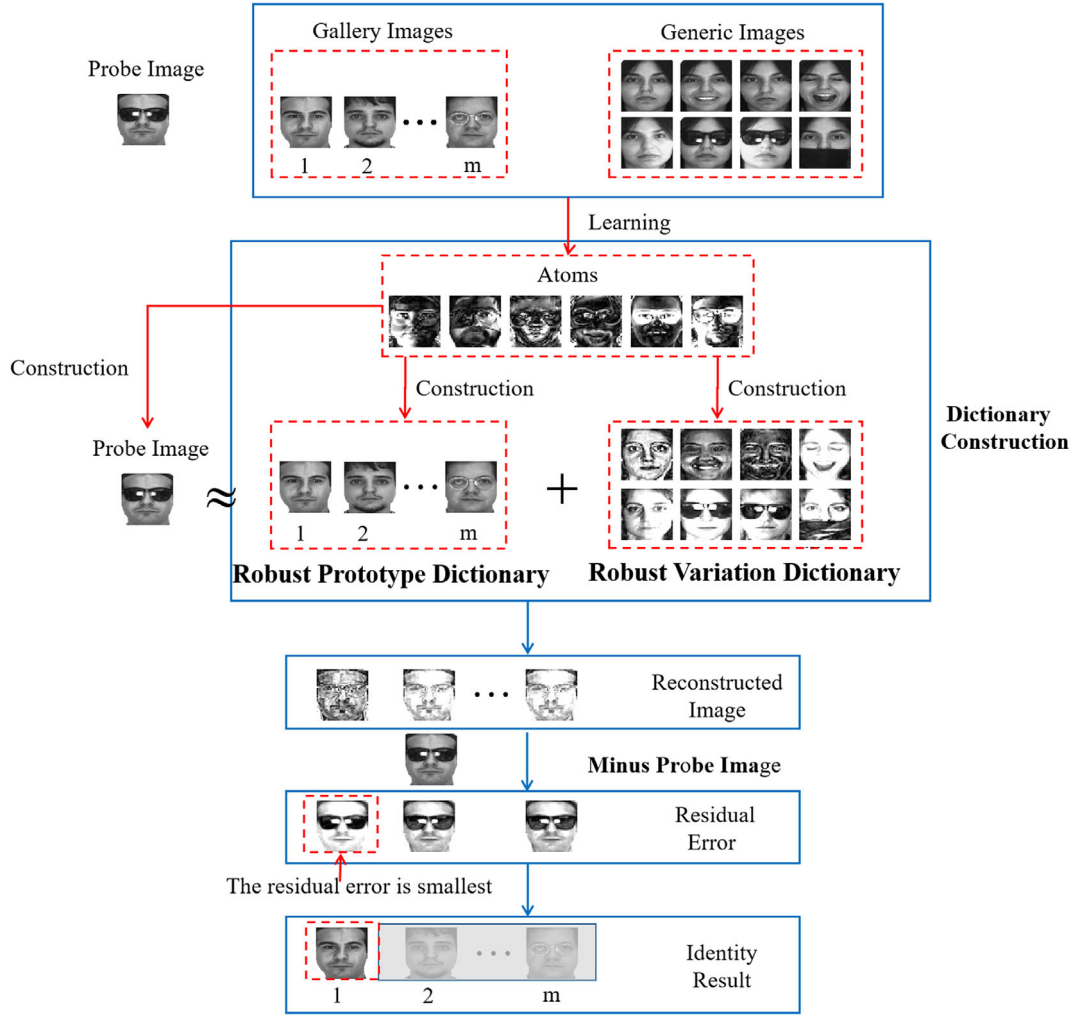


FIGURE 1 The sketch map of the RPRV method

In this step, due to the high dimension of each column of the matrix $\mathbf{X}$, it is time consuming to learn a dictionary $\mathbf{D}$ directly. Therefore, we use the skinny singular value decomposition (skinny SVD) method [41] to reduce dimensions as given by

$$\mathbf{X} = \mathbf{U}_r \sum_r \mathbf{H}'_r, \quad (4)$$

where $\sum_r = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ is a diagonal matrix with r positive singular values in a descending order, $\mathbf{U}_r \in \mathbb{R}^{d \times r}$ and $\mathbf{H}_r \in \mathbb{R}^{(m+n) \times r}$ are column-wise orthogonal matrices satisfying $\mathbf{U}'_r \mathbf{U}_r = \mathbf{H}'_r \mathbf{H}_r = \mathbf{I}$.

Let $\tilde{\mathbf{X}} = \sum_r \mathbf{H}'_r \in \mathbb{R}^{r \times (m+n)}$ ($r \leq (m+n) \ll d$) be a matrix with the reduced dimension and we learn dictionary $\tilde{\mathbf{D}}$ to represent $\tilde{\mathbf{X}}$ under sparse constraints. Specifically, the following objective function is used for minimization problem given by

$$\begin{aligned} \min_{\tilde{\mathbf{D}}, \tilde{\mathbf{W}}} & \|\tilde{\mathbf{X}} - \tilde{\mathbf{D}}\tilde{\mathbf{W}}\|_F^2 + \gamma \|\tilde{\mathbf{W}}\|_1 \\ \text{s.t.} & \tilde{\mathbf{d}}'_i \tilde{\mathbf{d}}_i = 1, \forall i. \end{aligned} \quad (5)$$

Equation (5) is a joint optimization problem of the dictionary $\tilde{\mathbf{D}}$ and the coefficient matrix $\tilde{\mathbf{W}}$. Although it is not jointly convex with respect to $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{W}}$, it is convex with respect to each of them when the other one is fixed. Therefore, the coefficient matrix $\tilde{\mathbf{W}}$ is optimized using the basis pursuit denoising (BPDN)-homotopy algorithm in [42], when the dictionary $\tilde{\mathbf{D}}$ is fixed.

Then the dictionary $\tilde{\mathbf{D}}$ is optimized using metaface learning algorithm in [43], when the coefficient matrix $\tilde{\mathbf{W}}$ is fixed.

$$J_{\tilde{\mathbf{D}}} = \arg \min_{\tilde{\mathbf{D}}} \left\{ \|\tilde{\mathbf{X}} - \tilde{\mathbf{D}}\tilde{\mathbf{W}}\|_F^2 \right\} \quad \text{s.t.} \tilde{\mathbf{d}}_j^T \tilde{\mathbf{d}}_j = 1, \forall j, \quad (6)$$

where dictionary matrix $\tilde{\mathbf{D}} = [\tilde{\mathbf{d}}_1, \tilde{\mathbf{d}}_2, \dots, \tilde{\mathbf{d}}_k]$, $\tilde{\mathbf{d}}_j \in \mathbb{R}^{r \times 1}$, and coefficient matrix $\tilde{\mathbf{W}} = [\tilde{\mathbf{w}}_1; \tilde{\mathbf{w}}_2; \dots, \tilde{\mathbf{w}}_k] \in \mathbb{R}^{k \times (m+n)}$, $\tilde{\mathbf{w}}_j \in \mathbb{R}^{1 \times (m+n)}$. In this phase, we update $\tilde{\mathbf{d}}_j$ one by one by fixing the other columns in $\tilde{\mathbf{D}}$, then Equation (6) is transformed into

$$J_{\tilde{\mathbf{d}}_j} = \arg \min_{\tilde{\mathbf{d}}_j} \left\| \mathbf{Y} - \tilde{\mathbf{d}}_j \tilde{\mathbf{w}}_j \right\|_F^2 \quad \text{s.t.} \tilde{\mathbf{d}}_j^T \tilde{\mathbf{d}}_j = 1, \forall j, \quad (7)$$

where $\mathbf{Y} = \tilde{\mathbf{X}} - \sum_{i \neq j} \tilde{\mathbf{d}}_i \tilde{\mathbf{w}}_i$.

ALGORITHM 1 Dictionary Learning algorithm

Input: gallery data matrix $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_m] \in \mathbb{R}^{d \times m}$ generic data matrix $\hat{\mathbf{V}} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] \in \mathbb{R}^{d \times n}$

- 1: Gallery data matrix $\mathbf{G}$ and variation matrix $\hat{\mathbf{V}}$ are merged into the matrix $\mathbf{X} \in \mathbb{R}^{d \times (m+n)}$
- 2: the dimension of samples is reduced by skinny SVD
 $\mathbf{X} = \mathbf{U}_r \sum_r \mathbf{H}'_r$, low-dimension samples matrix $\tilde{\mathbf{X}} = \sum_r \mathbf{H}'_r \in \mathbb{R}^{r \times (m+n)}$ is obtained
- 3: dictionary learning $\tilde{\mathbf{X}} = \tilde{\mathbf{D}} \tilde{\mathbf{W}}^T$
- 4: $\tilde{\mathbf{D}}$ is initialized as a random unit vector per column
- 5: **while do**
- 6: Fix $\tilde{\mathbf{D}}$, update $\tilde{\mathbf{W}}$ by BPDN homotopy method
- 7: Fix $\tilde{\mathbf{W}}$, update $\tilde{\mathbf{D}}$
- 8: **for** $j = 1$ to k **do**
- 9: Fix $\tilde{\mathbf{d}}_j, i \neq j$, update $\tilde{\mathbf{d}}_j$ using Equation (8)
- 10: **end for**
- 11: **end while**
- 12: $\mathbf{D} = \mathbf{U}_r \tilde{\mathbf{D}}, \mathbf{W} = \tilde{\mathbf{W}}$

Output: dictionary matrix $\mathbf{D}$ coefficient matrix $\mathbf{W}$

Then, let $J_{\mathbf{d}_j} = 0$, we get the solution as follows:

$$\tilde{\mathbf{d}}_j = \mathbf{Y} \tilde{\mathbf{w}}_j^T / \|\mathbf{Y} \tilde{\mathbf{w}}_j^T\|_2. \quad (8)$$

After obtaining $\tilde{\mathbf{D}}$ and $\mathbf{W}$, the dictionary $\mathbf{D}$ and the sparse coefficient matrix $\mathbf{Z}$ in Equation (3) can be obtained by

$$\begin{cases} \mathbf{D} = \mathbf{U}_r \tilde{\mathbf{D}} \\ \mathbf{W} = \tilde{\mathbf{W}} \end{cases}. \quad (9)$$

The summary dictionary learning algorithm is shown in Algorithm 1.

2.2 | Dictionary construction

The prototype dictionary and variation dictionary are constructed in this subsection. First, the coefficient matrix $\mathbf{W}$ in Section 2.1 can be written as $\mathbf{W} = [\mathbf{P}, \mathbf{Q}]$, where $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m] \in \mathbb{R}^{k \times m}$, $\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n] \in \mathbb{R}^{k \times n}$. For variation matrix $\hat{\mathbf{V}} \in \mathbb{R}^{d \times n}$, let $\mathbf{v}_i, i = 1, 2, \dots, n$ be column i of the matrix $\hat{\mathbf{V}}$, $\mathbf{v}_i$ can be represented as follows:

$$\mathbf{v}_i \approx \mathbf{D} \mathbf{q}_i = \sum_{j=1}^k \mathbf{d}_j q_{i,j}. \quad (10)$$

Let $\tilde{\mathbf{v}}_{i,j} = \mathbf{d}_j q_{i,j}$, Equation (10) can be written as the summation of k components, given by

$$\mathbf{v}_i \approx \sum_{j=1}^k \tilde{\mathbf{v}}_{i,j}. \quad (11)$$

Then we use $\tilde{\mathbf{v}}_{i,j}$ and its label information to calculate the fisher ratio of each column $\mathbf{d}_j$, the bigger the fisher ratio of column $\mathbf{d}_j$, the more discriminative to separate the class of samples.

We denote the column mean of $\tilde{\mathbf{v}}_{i,j}, i = 1, 2, \dots, n$ by $\bar{\mathbf{v}}_j$ and the column mean of $\mathbf{v}_{i,j}, i = 1, 2, \dots, n_c$ belonging to class Ω_c by $\bar{\mathbf{v}}_j^c$, where $\Omega_c, c = 1, 2, \dots, C$ denotes the class of the samples $\mathbf{v}_i$, n_c is the number of samples $\mathbf{v}_i$ belonging to class Ω_c . The fisher ratio [44] of feature $\mathbf{v}_{i,j}$ is written as follows:

$$f_v(j) = \frac{\sigma_b}{\sigma_w} = \frac{\sum_{c=1}^C (\bar{\mathbf{v}}_j^c - \bar{\mathbf{v}}_j)^2}{\sum_{c=1}^C \sum_{\mathbf{v}_i \in \Omega_c} (\tilde{\mathbf{v}}_{i,j} - \bar{\mathbf{v}}_j^c)^2}, \quad (12)$$

where σ_b denotes inter-class scatter, σ_w denotes intra-class scatter.

Each gallery sample can be represented as follows:

$$\mathbf{g}_i \approx \mathbf{D} \cdot \mathbf{p}_i = \sum_{j=1}^k \mathbf{d}_j p_{i,j}. \quad (13)$$

Let $\tilde{\mathbf{g}}_{i,j} = \mathbf{d}_j p_{i,j}$, Equation (13) can be written as the summation of k components, given by

$$\mathbf{g}_i \approx \sum_{j=1}^k \tilde{\mathbf{g}}_{i,j}. \quad (14)$$

Because each class has only single sample, there is no intra-class scatter information. We calculate the inter-class scatter

$$f_g(j) = \sum_{i=1}^m (\tilde{\mathbf{g}}_{i,j} - \bar{\mathbf{g}}_j)^2, \quad (15)$$

where $\bar{\mathbf{g}}_j = \sum_{i=1}^m \tilde{\mathbf{g}}_{i,j} / m$.

In the P+V model, we hope to increase the inter-class scatter of the prototype dictionary, that is, we need $f_g(j)$ to be as large as possible. At the same time, we hope to reduce the inter-class scatter and increase the intra-class scatter in the variation dictionary, that is, we need $f_v(j)$ to be as small as possible. Therefore, we construct formula (16); the smaller $f(j)$ is, the better the prototype dictionary and the variation dictionary.

$$f(j) = f_v(j) - f_g(j). \quad (16)$$

We sort $f(j)$ in an ascending order, and obtain the position sequence $\text{ind}(j), j = 1, 2, \dots, k$. We choose the sum of the first $k/2$ as features, given by

$$\tilde{\mathbf{v}}_i = \sum_{j=\text{ind}(1)}^{\text{ind}(k/2)} \tilde{\mathbf{v}}_{i,j}, i = 1, 2, \dots, n. \quad (17)$$

Then, the variation dictionary is obtained as follows:

$$\tilde{\mathbf{V}} = [\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_n]. \quad (18)$$

We choose the sum of the first $k/2$ position sequence $\text{ind}(j)$ as features, given by

$$\tilde{\mathbf{g}}_i = \sum_{j=\text{ind}(1)}^{\text{ind}(k/2)} \tilde{\mathbf{g}}_{i,j}, i = 1, 2, \dots, m. \quad (19)$$

ALGORITHM 2 Dictionary Construction**Input:** Variation dictionary $\mathbf{D}$, Sparse coefficient matrix $\mathbf{W}$

- 1: We divide matrix $\mathbf{W}$ into matrix $\mathbf{P}$ and matrix $\mathbf{Q}$
- 2: For each generic sample $\hat{\mathbf{v}}_i, i = 1, 2, \dots, n$, construct components $\tilde{\mathbf{v}}_{i,j} = \mathbf{d}_j q_{i,j}$
- 3: For components $\tilde{\mathbf{v}}_{i,j}, i = 1, 2, \dots, n$, obtain the fisher ratio $f_v(j)$
- 4: For each gallery sample $\mathbf{g}_i, i = 1, 2, \dots, m$, construct components $\tilde{\mathbf{g}}_{i,j} = \mathbf{d}_j p_{i,j}$
- 5: For components $\tilde{\mathbf{g}}_{i,j}, i = 1, 2, \dots, m$, obtain the inter-class scatter $f_g(j)$
- 6: $f(j) = f_v(j) - f_g(j)$
- 7: We sort $f(j)$ in an ascending order, and obtain the position sequence $\text{ind}(j), j = 1, 2, \dots, k$
- 8: Obtain variation dictionary $\tilde{\mathbf{V}}$
- 9: Obtain prototype dictionary $\tilde{\mathbf{G}}$

Output: Variation dictionary $\tilde{\mathbf{V}}$ Prototype dictionary $\tilde{\mathbf{G}}$

Then, the prototype dictionary $\tilde{\mathbf{G}}$ is obtained as follows:

$$\tilde{\mathbf{G}} = [\tilde{\mathbf{g}}_1, \dots, \tilde{\mathbf{g}}_m]. \quad (20)$$

The robust dictionaries construction algorithm is shown in Algorithm 2.

2.3 | Face recognition

In this section, we introduce sparse representation classification method based on a prototype plus variation (P+V) model for face recognition using our proposed RPRV method.

First, according to the dictionary $\mathbf{D}$ obtained in Section 2.1, the dictionary coefficient $\mathbf{w} \in \mathbb{R}^{d \times 1}$, ($\mathbf{y} = \mathbf{D} \cdot \mathbf{w}$) of the probe sample is obtained using the basis pursuit denoising (BPDN)-homotopy algorithm [42].

According to the $\text{ind}(j), j = 1, 2, \dots, k$ obtained in Section 2.2, we choose the sum of the first $k/2$ as a new sample, given by

$$\tilde{\mathbf{y}} = \sum_{j=\text{ind}(1)}^{\text{ind}(k/2)} \mathbf{d}_j w_j. \quad (21)$$

Let $\tilde{\mathbf{G}} \in \mathbb{R}^{d \times m}$ be a prototype dictionary matrix, which are composed of gallery samples; the column of $\tilde{\mathbf{G}}$ denotes the single sample of each person in the gallery. Let $\tilde{\mathbf{V}} \in \mathbb{R}^{d \times n}$ be a variation dictionary matrix, which represents the intra-class variation per person, such as expression, illumination, pose and occlusion. Then, the linear representation of a probe sample $\tilde{\mathbf{y}}$ is rewritten as

$$\tilde{\mathbf{y}} = \tilde{\mathbf{G}}\alpha + \tilde{\mathbf{V}}\beta + \mathbf{z}, \quad (22)$$

where $\alpha \in \mathbb{R}^{m \times 1}$ is the corresponding coefficient of $\tilde{\mathbf{G}}$, the column of prototype dictionary denotes the different subjects, the coefficients are sparse, most coefficients among α are 0, $\beta \in$

$\mathbb{R}^{n \times 1}$ is the corresponding coefficient of $\tilde{\mathbf{V}}$, $\mathbf{z} \in \mathbb{R}^{d \times 1}$ denotes the noise term.

Normalize the columns of $\tilde{\mathbf{G}}$ and $\tilde{\mathbf{V}}$ to have unit ℓ_2 -norm, and solve the ℓ_1 -minimization problem.

$$\begin{bmatrix} \alpha^* \\ \beta^* \end{bmatrix} = \arg \min \left\| \tilde{\mathbf{y}} - [\tilde{\mathbf{G}} \quad \tilde{\mathbf{V}}] \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \right\|_2 + \lambda \left\| \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \right\|_1. \quad (23)$$

The probe sample $\tilde{\mathbf{y}}$ can be classified according to the residual $r_i(\tilde{\mathbf{y}})$.

$$r_i(\tilde{\mathbf{y}}) = \left\| \tilde{\mathbf{y}} - [\tilde{\mathbf{g}}_i \quad \tilde{\mathbf{V}}] \begin{bmatrix} \alpha_i^* \\ \beta^* \end{bmatrix} \right\|_2^2, i = 1, 2, \dots, m, \quad (24)$$

where $\delta_i(\alpha^*)$ is a vector; the entries belonging to the class Ω_i are the corresponding values in α^* , and the other entries are 0.

The identity of probe sample $\tilde{\mathbf{y}}$ can be obtained according to the smallest reconstruction residual.

$$\text{Identity}(\tilde{\mathbf{y}}) = \arg \min_i r_i(\tilde{\mathbf{y}}). \quad (25)$$

3 | EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first introduce standard public face datasets, including AR, Extended YaleB (E-YaleB), CMU PIE, CAS-PEAL, the unconstrained labeled Faces in the Wild (LFW), Celebfaces Attribute (CelebA) datasets and VggFace (Visual Geometry Group Face) in Section 3.1. Then, in Section 3.2, we use four experiments to evaluate the performance of the proposed RPRV method in constrained environments. In Section 3.3, the performance of the proposed RPRV method is evaluated using three datasets in unconstrained environments. Finally, in Section 3.4, several experiments are conducted to evaluate the computational complexity using the proposed RPRV method and some SSPP face recognition methods.

3.1 | Datasets

In this subsection, we give a brief introduction of the datasets used in the experiments.

The AR dataset [45] is composed of S-I and S-II subsets, which are collected in different periods, including over 4000 samples of 126 subjects, and there are 13 samples per subject in S-I and S-II. Moreover, various facial variations exist over these samples, such as smiling, angering, lighting, wearing sunglasses and so on.

The Extended YaleB (E-YaleB) dataset [46] contains 2414 images of 38 subjects under different illumination conditions, which is separated into five subsets according to the illumination angle, such as 12° , 25° , 50° , 77° ; each subset contains 7, 12, 14, 12 and 19 samples per subject, respectively.



FIGURE 2 The samples of AR dataset including expression, illumination, sunglasses, scarf and disguise

The CMU PIE dataset [47] comes from Carnegie Mellon University, which consists of 41,368 images of 68 subjects, and each subject has 13 different poses, 43 different illumination conditions, and with 4 different expressions. This database is referred to as the CMU PIE dataset.

The CAS-PEAL dataset [48] is from Chinese Academy of Sciences (CAS), which provides large-scale face images with different sources of variations, especially pose, expression, accessories, and lighting (PEAL); this dataset contains 99,594 images of 1040 subjects (595 males and 445 females).

The labeled faces in the wild (LFW) dataset [49] include 13,000 images and 5749 subjects. The face images are collected in an unconstrained environment; the variation degree of the image is larger than that in the constrained environments, so the SSPP task for this dataset is more difficult.

CelebFaces Attribute (CelebA) dataset [50] contains 10,177 celebrity identity, including 202,599 face images. Each image contains face tagging box, five face feature point coordinates (two eyes, nose tip, two mouth corners) and 40 attribute markers. CelebA is provided by the Chinese University of Hong Kong. It can be widely used for the face identification training and testing.

VggFace (Visual Geometry Group Face) dataset [51] that crawls images of celebrities on the Internet, includes 2622 celebrities; each celebrity has 1000 images. The dataset is designed to have no overlap with popular facial recognition benchmark datasets, including labeled faces in the wild (LFW) and youtube faces (YTF) dataset.

3.2 | Performance of RPRV in constrained environments

In this subsection, we verify the effectiveness of the proposed RPRV method using four public datasets in a constrained environment. For the AR dataset, 50 subjects are selected from S-I as the generic set; each person contains one frontal face image without illumination and expression variations and other 12 images include facial variations of smiling, angering, lighting, wearing sunglasses and scarf as shown in Figure 2, for example. Then we select another 50 subjects for evaluation. For each person, the frontal face image without illumination and expression

changes is used as the gallery set, and the rest of 12 images are used as the probe set. Moreover, the samples from S-II are used as the probe set.

For the E-YaleB dataset, the samples under various lighting conditions are shown in Figure 3. 18 subjects are selected as the generic set, and the rest of the 20 subjects for evaluation. For each person, the first image in Subset 1 is used as the gallery set, and the rest of subsets are separately used as the probe sets.

For the CMU PIE dataset, the samples under various lighting and pose conditions are shown in Figure 4. we randomly select 28 subjects used as the generic set, and the rest of 40 subjects for evaluation. Usually, the experiment uses the subsets of five poses: C27, C05, C07, C09 and C29. Where the C27 subset is frontal pose, which is used as the gallery set, and the other four subsets as the probe sets.

For the CAS-PEAL dataset, we randomly select 100 subjects used as the generic set, each person contains one frontal face image from normal subset, three face images with glasses and three faces with hats from accessory subset as shown in Figure 5. We select the rest of the 50 subjects for evaluation; for each person, the normal samples are used as the gallery set, and 6 accessory samples (glasses and hats) are used as probe sets.

For the above four datasets, we resize all of the image to 48×48 , and reshape a column vector as face features, then, we perform the recognition. In the process of the recognition, the analysis of the regularization parameters λ in formula (23) are shown in Figure 6. Generally, in references such as CPL [35], SGL [40] etc., the value range of the regularization parameters λ is usually between 0.001 and 0.1. In Figure 6, we can see that with the increase of the regularization parameters λ , the recognition accuracy decreases. Therefore, when the value of regularization parameter λ is 0.001, the recognition accuracy is the highest.

Then, we verify the number of atoms indexes in the reconstructed samples. We take the difference values between the $f_v(j)$ and $f_g(j)$ in ascending order. And we select the first 0.25k, 0.5k and 0.75k in order as the index for the reconstruct samples. The experimental results are shown in Figure 7. It can be seen that the experimental effect is the best when 0.5k indexes are taken. Therefore, the first 0.5k is selected as the index for the reconstructed samples in this experiment.



FIGURE 3 The samples of E-YaleB dataset under various lighting conditions



FIGURE 4 The samples of CMU PIE dataset under various lighting and pose conditions

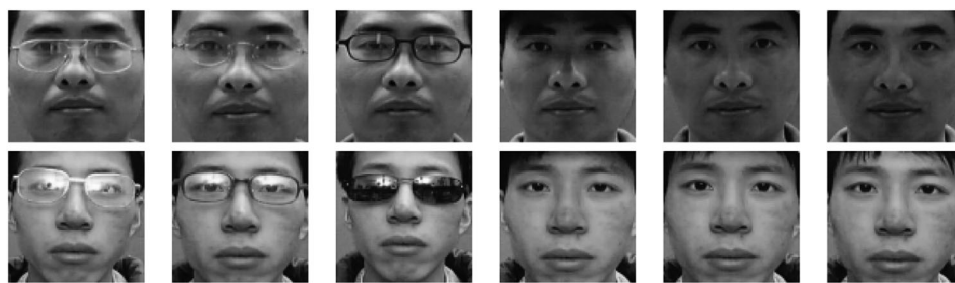


FIGURE 5 The samples of CAS-PEAL dataset under various glasses and hats conditions

We list six representative comparison methods, including the SRC [32] method, and five generic learning methods, including ESRC [33], SSRC [36], SVDL [34], CPL [35] and SGL [40]. The regularization parameters λ is fixed as 0.001. The comparative experiment results of different methods as shown in Table 1. Our method is better than other generic learning methods, and the experimental results of these comparison methods are cited from [40]. In addition, the performance of our method is always

better than SGL on four probe datasets. SGL only improves the intra-class scatter in variation dictionary, while our RPRV method not only improves the inter-class discriminative ability in the robust prototype dictionary, but also increases the intra-class scatter in the robust variation dictionary, thus the robustness of our RPRV method is better than SGL. ESRC and SSRC obtain the variation dictionary by the difference between the corresponding pixels in the image, the simple construction of

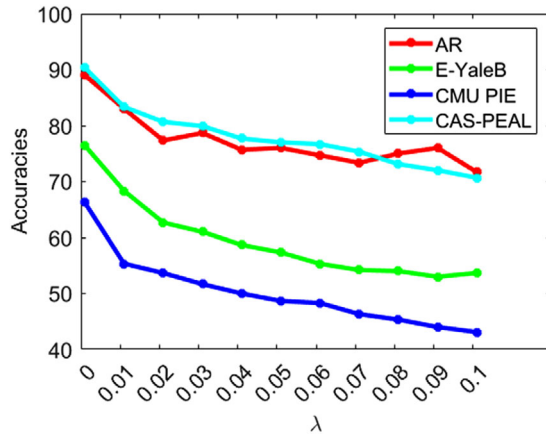


FIGURE 6 The sensitivity analysis of the regularization parameters λ on recognition performance

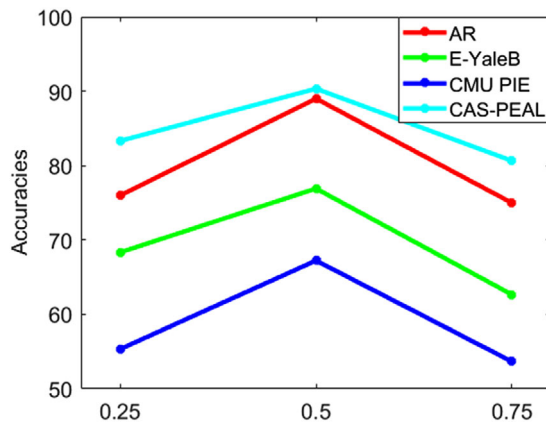


FIGURE 7 The recognition performance versus the number of atoms indexes in the reconstructed samples

TABLE 1 Accuracies (%) on the AR, E-YALEB, CMU PIE and CAS-PEAL datasets

Probe set		SRC[32]	ESRC[33]	SSRC[36]	SVDL[34]	CPL[35]	SGL[40]	RPRV
AR	S-1	56.83	84.58	86.36	88.14	87.08	92.55	95.16
	S-2	36.72	67.33	70.78	73.66	70.94	77.89	83.23
	Average	46.78	75.96	78.57	80.9	79.01	85.22	89.20
E-YaleB	Subset2	96.67	99.83	100.00	100.00	100.00	100.00	100.00
	Subset3	56.67	95.33	97.33	98.58	98.33	100.00	100.00
	Subset4	15.00	61.43	69.29	71.07	70.71	78.89	79.50
	Subset5	7.63	16.05	18.95	20.79	20.37	24.74	28.17
	Average	43.99	68.16	71.39	72.61	72.35	75.59	76.92
CMU PIE	C27 subset	44.44	73.89	75.94	75.00	76.67	80.00	89.04
	C05 subset	29.50	56.75	64.50	64.75	64.25	70.50	70.67
	C07 subset	32.00	48.25	55.00	55.75	55.50	60.50	61.24
	C09 subset	31.50	54.25	58.50	56.75	58.25	62.00	62.08
	C29 subset	15.50	34.00	39.75	40.50	43.75	52.50	53.1
CAS-PEAL	Average	30.59	53.43	58.54	58.55	59.68	65.10	67.23
	Accessory:Glasses	94.00	96.00	97.33	98.00	97.33	98.67	99.3
	Accessory:Hats	52.67	64.00	68.00	68.67	68.67	76.67	81.36
	Average	73.34	80.00	82.67	83.34	83.00	87.67	90.33

variation dictionary leads to the low performance of the algorithm. CPL is an improved method of SVDL; this method needs lots of generic samples to contribute the variation dictionary, however, the number of samples is usually limited. The accuracy of the proposed method is better than the other six methods. Therefore, the experiment results show that our RPRV method is effective and superior in the SSPP face recognition problem.

3.3 | Performance of RPRV in unconstrained environments

3.3.1 | Evaluation on LFW dataset

In this subsection, we verify our proposed algorithm on the unconstrained LFW dataset, the samples of LFW as shown in Figure 8; these samples have large variations in expressions, poses, illuminations.

We selected samples with more than 9 images per person and selected 158 subjects to evaluate in the experiment. We select 108 subjects as the generic set, and the other 50 subjects are selected for evaluation. For each subject, we randomly select one image as the gallery set, and select the rest of the images as the probe set. As shown in Table 2, we use three recent deep learning method, VggFace [51], CosFace [52] and ArcFace [53], to extract features to evaluate the effectiveness of the proposed method. According to the experiment setting of comparative literature, 4096 dimensional vector of the first full connection layer in VggFace are extracted to perform the SSPP face recognition, and the features dimensions of CosFace and ArcFace are both 512. Our RPRV is compared with SSRC, S^3 RC and SGL.



FIGURE 8 The samples of LFW dataset under unconstrained conditions

TABLE 2 Accuracies (%) and standard errors (%) of different methods on the LFW dataset, with Raw Pixels, VGG-Face, CosFace and ArcFace feature

Method	Raw Pixels	VGG-Face	CosFace	ArcFace
SSRC[36]	20.75±3.24	89.31±2.49	90.65±1.5	91.16±1.3
S ³ RC[39]	23.82±2.9	90.51±2.42	92.8±1.45	93.35±1.42
SGL[40]	24.35±3.82	93.11±2.06	93.65±1.61	93.74±1.24
RPRV	26.23±2.53	96.88±2.31	97.1±1.36	97.54±1.08

We simulate the above 3 comparison methods. The experiment is repeated 30 times. From Table 2 we know that the ArcFace feature is better than the other three features, and the accuracy of the proposed method is better than the other 3 comparing methods. Therefore, the experiment shows that our method is the best in recognition performance.

3.3.2 | Evaluation on CelebA dataset

In this subsection, we verify our algorithm on the unconstrained CelebA dataset, the samples of CelebA as shown in Figure 9.

In the CelebA dataset, each subject has a different number of samples. There are 2360 subjects and each subject more than

30 samples. We selected 100 subjects to evaluate the experiment, and the other subjects as the generic set for evaluation. We use ArcFace [53] method to extract 512 dimensional features to evaluate the effectiveness of the proposed method. As shown in Table 3, we simulate the SSRC[36], S³RC[39] and SGL[40] comparison methods. The experiment is repeated 30 times. We see that as the number of subjects in the generic set increases, the recognition accuracy is increasing.

3.3.3 | Evaluation on VggFace dataset

In this subsection, we verify our algorithm on the unconstrained VggFace dataset; the samples of VggFace as shown in Figure 10.

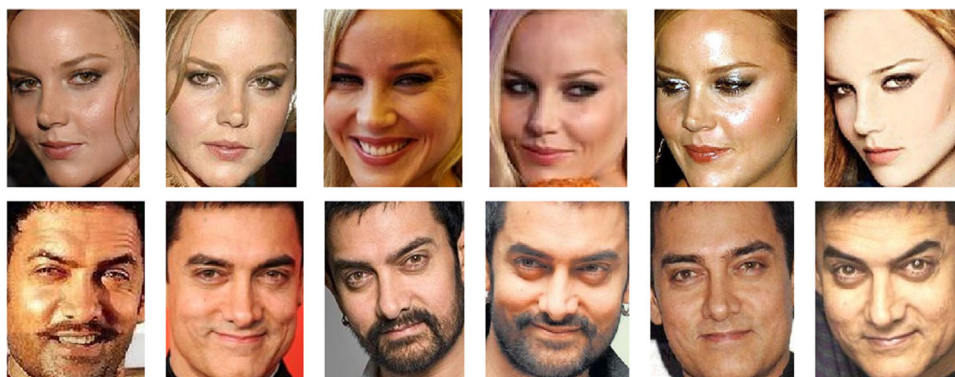
In the VggFace dataset, each subject has a different number of samples. There are 2360 subjects and each subject has more than 30 samples. We selected 100 subjects as the generic set, and the other subjects are selected for evaluation. We extract ArcFace [53] features to evaluate the effectiveness of the proposed method. The experiment is repeated 30 times. As shown in Table 4, we see that as the number of subjects in the gallery set increases, the recognition accuracy of our method and SSRC[36], S³RC[39] and SGL[40] comparison methods decreases.



FIGURE 9 The samples of CelebA dataset under unconstrained conditions

TABLE 3 Recognition accuracies of compared methods on CelebA as the number of subjects in the generic set increases

Methods	Generic subjects number					
	30	50	100	150	200	250
SSRC[36]	78.98±1.66	79.01±1.82	79.14±1.42	79.1±1.9	79.22±1.3	79.21±1.77
S ³ RC[39]	81.23±1.38	81.45±1.4	82.52±1.75	82.67±1.62	82.95±1.63	83.45±1.51
SGL[40]	80.42±1.18	81.21±1.08	81.23±1.36	82.4±1.67	83.1±1.82	83.48±1.42
RPRV	85.78±1.06	85.82±1.14	86.3 ±1.32	85.82±1.17	86.53±1.43	87.4±1.05

**FIGURE 10** The samples of VggFace dataset under unconstrained conditions**TABLE 4** Recognition accuracies of compared methods on VggFace as the number of subjects in the gallery set increases

Methods	Gallery subjects number					
	100	200	300	400	500	600
SSRC[36]	78.31±1.67	76.42±1.46	75.01±1.33	74.96±1.62	74.38±1.49	73.52±2.13
S ³ RC[39]	79.24±1.56	78.24±1.82	76.57±1.52	76.3±1.75	75.74±1.21	75.32±1.79
SGL[40]	81.58±1.26	78.11±2.2	76.41±1.63	75.23±1.37	74.53±1.88	74.1±1.57
RPRV	86.09±1.41	83.47±1.47	80.22±1.07	80.3±1.32	79.43±1.57	78.52±1.36

TABLE 5 Comparison of recognition accuracies of state of the art SSPP methods on LFW

Method	LGR[38]	JCR-ACF[37]	LRGR-BSS[54]	RHDA[55]
Accuracy(%)	84.4	86.0	86.1	87.56
Method	S ³ RC[39]	SGL[40]	CJR-RACF[56]	RPRV
Accuracy(%)	92.57	93.11	95.5	96.88

3.4 | Comparison with state-of-the-art method

Seven state-of-the-art SSPP face recognition methods, that is, LGR [38], JCR-ACF [37], S³RC [39], LRGR-BSS [54], RHDA [55], SGL [40] and CJR-RACF [56] are selected for comparison with our proposed method. The comparative experiment results of the LFW dataset are shown in Table 5. The proposed

method is much better than the comparing methods. The comparison method only considers the prototype matrix or variation matrix, but our dictionary construction method considers both prototype matrix and variation matrix. Experiment results show that our proposed method is very effective for the SSPP problem with the extreme variation on unconstrained environment. In addition, the proposed method can directly use more robust deep learning features for recognition tasks.

3.5 | Complexity analysis

Complexity analysis is a vital factor in lots of realtime face recognition applications. In this subsection, we choose four generic learning SSPP face recognition methods for comparison; SRC[32] and SSRC [36] are the basic methods of generic learning, S³RC [39] and SGL [40] are the global generic learning methods. The time complexity of our RPRV method is

TABLE 6 Average time required by techniques to classify a probe sample on the LFW dataset

Method	SRC [32]	SSRC [36]	S ³ RC [39]	SGL [40]	RPRV
Time (s)	0.04	0.07	0.3	0.11	0.09

$O(d^2(m+n+k))$, where d denotes the face feature dimension, m is classes number in gallery set. In the SSPP problem, each class only contains a single sample, and the number of gallery samples is m . n is the number of samples in the auxiliary generic set. The atoms in the dictionary are learned by using these generic samples; the number of atoms is k , $k \leq n$. The complexity of SRC is $O(d^2n)$. The complexity of SSRC is $O(d^2(m+n))$. The complexity of S³RC is $O(2d^2(m+n))$. The complexity of SGL is $O(d^2(m+3n))$.

The average test time required to classify a probe sample on the LFW dataset is shown in Table 6. Although the computational complexity of our RPRV is larger than that of SRC [32] and SSRC 36, the performance of RPRV is much better than them. The computational complexity of RPRV is smaller than that of S³RC [39] and SGL [40], and its performance is better than them. In a summary, RPRV has time efficiency and recognition accuracy advantages in practical applications.

4 | CONCLUSION

The proposed RPRV method in this paper constructs the robust prototype dictionary and the robust variation dictionary. Using the robust prototype dictionary and robust variation dictionary, the proposed RPRV method is fit for the SSPP face recognition problem in unconstrained environment. First, a set of atoms is obtained by dictionary learning method using gallery images and generic images; some effective atoms are selected by the proposed function index method. Then, the robust prototype dictionary (RP) and the robust variation dictionary (RV) are constructed using these effective atoms. Finally, the constructed robust prototype dictionary (RP) and the constructed robust variation dictionary (RV) are used to recognize the probe sample, our RP+RV model can accurately recognize in unconstrained environment. Experimental results on several public face datasets show that our proposed RPRV method achieves higher accuracy in the SSPP problem, and the computational complexity of RPRV is not high in comparison methods. A more effective variation dictionary will be established in future research to improve the SSPP face recognition accuracy.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data derived from public domain resources.

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REFERENCES

- Guo, G., Zhang, N.: A survey on deep learning based face recognition. *Comput. Vis. Image Understand.* 189, 102805 (2019)
- Zhao, W., Chellappa, R., Phillips, P.J., et al.: Face recognition: A literature survey. *ACM Comput. Surv.* 35(4), 399–458 (2003)
- Du, H., Shi, H., Liu, Y., et al.: Towards NIR-VIS masked face recognition. *IEEE Signal Process. Lett.* 28, 768–772 (2021)
- Low, C.Y., Teoh, A.B.J., Park, J.: MIND-Net: A deep mutual information distillation network for realistic low-resolution face recognition. *IEEE Signal Process. Lett.* 28, 354–358 (2021)
- Karimi-Rouzbahani, H., Ramezani, F., Woolgar, A., et al.: Perceptual difficulty modulates the direction of information flow in familiar face recognition. *NeuroImage* 233, 117896 (2021)
- Wu, F., Jing, X.Y., Feng, Y., et al.: Spectrum-aware discriminative deep feature learning for multi-spectral face recognition. *Pattern Recognit.* 111, 107632 (2021)
- Tan, X., Chen, S., Zhou, Z.H., et al.: Face recognition from a single image per person: A survey. *Pattern Recognit.* 39(9), 1725–1745 (2006)
- Bashbaghi, S., Granger, E., Sabourin, R., et al.: Dynamic ensembles of exemplar-SVMs for still-to-video face recognition. *Pattern Recognit.* 69, 61–81 (2017)
- Mokhayeri, F., Granger, E.: A paired sparse representation model for robust face recognition from a single sample. *Pattern Recognit.* 100, 107129 (2020)
- Wagner, A., Wright, J., Ganesh, A., et al.: Toward a practical face recognition system: Robust alignment and illumination by sparse representation. *IEEE Trans. Pattern Anal. Mach. Intell.* 34(2), 372–386 (2012)
- Wu, F., Jing, X.Y., Feng, Y., et al.: Spectrum-aware discriminative deep feature learning for multi-spectral face recognition. *Pattern Recognit.* 111, 107632 (2021)
- Hu, C.H., Yu, J., Wu, F., et al.: Face illumination recovery for the deep learning system: Robust alignment and illumination by sparse representation. *Pattern Recognit.* 111, 107724 (2021)
- Plichoski, G.F., Chidambaram, C., Parpinelli, R.S.: A face recognition framework based on a pool of techniques and differential evolution. *Inf. Sci.* 543, 219–241 (2021)
- Li, Y., Huang, X., Zhao, G., et al.: Joint local and global information learning with single apex frame detection for micro-expression recognition. *IEEE Trans. Image Process.* 30, 249–263 (2021)
- Tang, Y., Zhang, X., Hu, X., et al.: Facial expression recognition using frequency neural network. *IEEE Trans. Image Process.* 30, 444–457 (2020)
- Chen, D., Song, P.: Dual-graph regularized discriminative transfer sparse coding for facial expression recognition. *Digital Signal Process.* 108, 102906 (2021)
- Cai, J., Han, H., Cui, J., et al.: Semi-supervised natural face de-occlusion. *IEEE Trans. Inf. Forensics Secur.* 16, 1044–1057 (2020)
- Hu, W., Hu, H.: Dual adversarial disentanglement and deep representation decorrelation for NIR-VIS face recognition. *IEEE Trans. Inf. Forensics Secur.* 16, 70–85 (2021)
- Gao, Q., Zhang, L., Zhang, D.: Face recognition using FLDA with single training image per person. *Appl. Math. Comput.* 205(2), 726–734 (2008)
- Deng, W., Hu, J., Wu, Z., et al.: From one to many: Pose-aware metric learning for single-sample face recognition. *Pattern Recognit.* 77, 426–437 (2018)
- Hu, C., Ye, M., Ji, S., et al.: A new face recognition method based on image decomposition for single sample per person problem. *Neurocomputing* 160, 287–299 (2015)

22. Plichoski, G.F., Chidambaram, C., Parpinelli, R.S.: A face recognition framework based on a pool of techniques and differential evolution. *Inf. Sci.* 543, 219–241 (2021)
23. Pang, M., Cheung, Y.M., Shi, Q., et al.: Iterative dynamic generic learning for face recognition from a contaminated single-sample per person. *IEEE Trans. Neural Networks Learn. Syst.* 32(4), 1560–1574 (2021)
24. Zhang, Z., Zhang, L., Zhang, M.: Dissimilarity-based nearest neighbor classifier for single-sample face recognition. *The Visual Comput.* 37(4), 673–684 (2021)
25. Zhu, P., Zhang, L., Hu, Q., et al.: Multi-scale patch based collaborative representation for face recognition with margin distribution optimization. In: *European Conference on Computer Vision*, pp. 822–835. Springer, Berlin (2012)
26. Zhang, L., Yang, M., Feng, X.: Sparse representation or collaborative representation: Which helps face recognition? *International Conference on Computer Vision*, pp. 471–478. IEEE, Piscataway (2011)
27. Yan, H., Lu, J., Zhou, X., et al.: Multi-feature multi-manifold learning for single-sample face recognition. *Neurocomputing* 143, 134–143 (2014)
28. Zhang, P., You, X., Ou, W., et al.: Sparse discriminative multi-manifold embedding for one-sample face identification. *Pattern Recognit.* 52, 249–259 (2016)
29. Adjabi, I., Ouahabi, A., Benzaoui, A., et al.: Multi-block color-binarized statistical images for single-sample face recognition. *Sensors* 21(3), 1–22 (2021)
30. Wei, C.P., Wang, Y.C.F.: Undersampled face recognition via robust auxiliary dictionary learning. *IEEE Trans. Image Process.* 24(6), 1722–1734 (2015)
31. Deng, W., Hu, J., Guo, J.: Face recognition via collaborative representation: Its discriminant nature and superposed representation. *IEEE Trans. Pattern Anal. Mach. Intell.* 40(10), 2513–2521 (2017)
32. Wright, J., Yang, A.Y., Ganesh, A., et al.: Robust face recognition via sparse representation. *IEEE Trans. Pattern Anal. Mach. Intell.* 31(2), 210–227 (2009)
33. Deng, W., Hu, J., Guo, J.: Extended SRC: Undersampled face recognition via intraclass variant dictionary. *IEEE Trans. Pattern Anal. Mach. Intell.* 34(9), 1864–1870 (2012)
34. Yang, M., Luc, V.G., Zhang, L.: Sparse variation dictionary learning for face recognition with a single training sample per person. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 689–696. IEEE, Piscataway (2013)
35. Ji, H.K., Sun, Q.S., Ji, Z.X., et al.: Collaborative probabilistic labels for face recognition from single sample per person. *Pattern Recognit.* 62, 125–134 (2017)
36. Deng, W., Hu, J.: In defense of sparsity based face recognition. In: *IEEE Conference on Computer Vision Pattern Recognition*, pp. 399–406. IEEE, Piscataway (2013)
37. Yang, M., Wang, X., Zeng, G., et al.: Joint and collaborative representation with local adaptive convolution feature for face recognition with single sample per person. *Pattern Recognit.* 66, 117–128 (2017)
38. Zhu, P., Yang, M., Zhang, L., et al.: Local generic representation for face recognition with single sample per person. In: *Asian Conference on Computer Vision*, pp. 34–50. Springer, Berlin Heidelberg (2014)
39. Gao, Y., Ma, J., Yuille, A.L.: Semi-supervised sparse representation based classification for face recognition with insufficient labeled samples. *IEEE Trans. Image Process.* 26(5), 2545–2560 (2017)
40. Pang, M., Cheung, Y.M., Wang, B., et al.: Synergistic generic learning for face recognition from a contaminated single sample per person. *IEEE Trans. Inf. Forensics Secur.* 15, 195–209 (2019)
41. Liu, G., Lin, Z., Yan, S., et al.: Robust recovery of subspace structures by low-rank representation. *IEEE Trans. Pattern Anal. Mach. Intell.* 35(1), 171–184 (2013)
42. Donoho, D.L., Tsai, Y.: Fast solution of l_1 -norm minimization problems when the solution may be sparse. *IEEE Trans. Inf. Forensics Secur.* 54(11), 4789–4812 (2008)
43. Yang, M., Zhang, L., Yang, J., et al.: Metaface learning for sparse representation based face recognition. In: *IEEE International Conference on Image Processing*, pp. 1601–1604. IEEE, Piscataway (2010)
44. Zhang, L., Zhu, P., Hu, Q., et al.: A linear subspace learning approach via sparse coding. In: *Proceedings of the International Conference on Computer Vision*, pp. 755–761. IEEE, Piscataway (2011)
45. Martinez, A., Benavente, R.: The AR face database. Technical Report. Computer Vision Center, Barcelona (1998)
46. Georgiades, A.S., Belhumeur, P.N., Kriegman, D.J.: From few to many: Illumination cone models for face recognition under variable lighting and pose. *IEEE Trans. Pattern Anal. Mach. Intell.* 23(6), 643–660 (2001)
47. Sim, T., Baker, S., Bsat, M.: The CMU pose, illumination, and expression (PIE) database. In: *IEEE International Conference on Automatic Face Gesture Recognition*, pp. 53–58. IEEE, Piscataway (2002)
48. Gao, W., Cao, B., Shan, S., et al.: The CAS-PEAL large-scale Chinese face database and baseline evaluations. *IEEE Trans. Syst. Man Cybern.* 38(1), 149–161 (2008)
49. Wolf, L., Hassner, T., Taigman, Y.: Effective unconstrained face recognition by combining multiple descriptors and learned background statistics. *IEEE Trans. Pattern Anal. Mach. Intell.* 33(10), 1978–1990 (2011)
50. Liu, Z., Luo, P., Wang, X., et al.: Deep learning face attributes in the wild. In: *International Conference on Computer Vision*, pp. 3730–3738. IEEE, Piscataway (2015)
51. Parkhi, O.M., Vedaldi, A., Zisserman, A.: Deep face recognition. In: *Proceedings of British Machine Vision Conference*, pp. 41.1–41.12. Springer, London (2015)
52. Wang, H., Wang, Y., Zhou, Z., et al.: CosFace: Large margin cosine loss for deep face recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5265–5274. IEEE, Piscataway (2018)
53. Deng, J., Guo, J., Xue, N., et al.: ArcFace: Additive angular margin loss for deep face recognition. In: *Proceedings of British Machine Vision Conference*, pp. 4685–4694. Springer, London (2019)
54. Liu, F., Ding, Y., Xu, F., et al.: Learning low-rank regularized generic representation with block-sparse structure for single sample face recognition. *IEEE Access* 7, 30573–30587 (2019)
55. Pang, M., Cheung, Y.M., Wang, B., et al.: Robust heterogeneous discriminative analysis for face recognition with single sample per person. *Pattern Recognit.* 89, 90–107 (2019)
56. Yang, M., Wen, W., Wang, X., et al.: Adaptive convolution local and global learning for class-level joint representation of facial recognition with a single sample per data subject. *IEEE Trans. Inf. Forensics Secur.* 15, 2469–2484 (2020)

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